

SHRIRAM CHANDRAN

Doctoral Researcher, Scalable Parallel Computing Laboratory, ETH Zürich

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Research Interests: heterogeneous accelerators · compiler systems · kernel optimization · performance modeling · AI/HPC systems

EDUCATION

ETH Zürich — Doctor of Sciences in Computer Science

2025–present

Advisor: Prof. Dr. Torsten Hoefler · Second Advisor: Prof. Dr. Markus Püschel

Zürich, Switzerland

ETH Zürich — MSc Computer Science

2022–2025

Thesis: Higher-Order Graph Databases · GPA: 5.35/6

Zürich, Switzerland

IIT Madras — BTech Computer Science and Engineering

2018–2022

Thesis: Program Analysis - StarPlat DSL · GPA: 9.04/10

Chennai, India

RESEARCH AND PROFESSIONAL EXPERIENCE

Doctoral Researcher, SPCL, ETH Zürich

2025–present

- **Teal:** Developing an architecture-aware tensor programming system for expert- and AI-generated high-performance kernels, exposing data movement, synchronization, and matrix-unit execution across heterogeneous accelerators.
- **CUTLASS Selector:** Built hardware-aware machine-learning models to select high-performance CUTLASS kernels without online benchmarking, trained on millions of measured configurations.
- **Fulcrum:** Designed a portable spatial-dataflow abstraction for programming heterogeneous matrix units across NVIDIA, AMD, Intel, and Arm architectures.
- **XLSDFT:** Optimized linear algebra kernels and analyzed multi-node performance for large-scale electronic-structure simulations on the Lineshine supercomputer.

Student Researcher, SPCL, ETH Zürich

2024–2025

- **HO-GDB:** Co-designed and implemented a higher-order graph database framework, including graph projections, analytical models, and OLTP/OLAP evaluation.

Student Researcher, Systems Group, ETH Zürich

2023–2024

- **Maximus:** Added GPU execution support to a heterogeneous query engine, including Arrow-compatible GPU data structures and TPC-H evaluation.

Undergraduate Researcher, IIT Madras

2021–2022

- **StarPlat DSL:** Implemented compiler-level optimizations for StarPlat, a graph processing DSL.
- **Stability Analysis:** Analyzed Bilu-Linial stability and approximation behavior for combinatorial optimization problems.

Software Engineering Intern, Rubrik

2021

- Built production infrastructure for collecting and analyzing backup-job performance statistics in distributed storage systems.

Software Engineering Intern, Godot Media

2020

- **Plagiarism Checker:** Built and parallelized an automated plagiarism-detection system for large text corpora against web documents.

PUBLICATIONS

What Bloats Your Floats? Right-Sizing Numerical Precision for Scientific Computing

Pratyai Mazumder, Alexandru Calotoiu, Marcin Copik, Yakup Koray Budanaz, **Shriram Chandran**, Torsten Hoefler

arXiv preprint, 2026 · [\[paper\]](#)

Hardware-Aware Features for CUTLASS Kernel Selection

Shriram Chandran, Dominic Rinderer, Yakup Budanaz, Alexandru Calotoiu, Marcin Copik, Torsten Hoefler

arXiv preprint, 2026 · [\[paper\]](#) · [\[code\]](#)

Extreme-Scale Linear-Scaling Kohn-Sham DFT at 100 Million Atoms: Bridging Quantum Simulations and Experiments

Qimen Xu, Yu Zhang, Dixing Ni, Lei Gao, Guangnan Feng, Qinrui Zheng, Jianting Liu, Haitian Lu, Zhaopeng Jia, Wei Xue, **Shriram Chandran**,

Torsten Hoefler, Haohuan Fu, Yutong Lu

arXiv preprint, 2026 · [\[paper\]](#) · [\[code\]](#)

Code Generation and Optimization for Different Targets for a Graph DSL

Ashwina Kumar, Naveen LS, **Shriram Chandran**, Hemesh DJ, Gudla Sanjay, Kovvuri Sravan Kumar Reddy, Nibedita Behera, Rupesh Nasre

Authorea preprint, 2025 · [\[paper\]](#)

The Data Must Flow (To Vector Processors): Searching Program Variants to Improve Compiler Auto-Vectorization Capabilities

Yakup Budanaz, **Shriram Chandran**, Alexandru Calotoiu, Torsten Hoefler

Technical report · [\[paper\]](#)

Psychologically Enhanced AI Agents

Maciej Besta, **Shriram Chandran**, Robert Gerstenberger, Mathis Lindner, Marcin Chrapek, Sebastian Hermann Martschat, Taraneh Ghandi, Patrick Iff, Hubert Niewiadomski, Piotr Nyczyk, Jürgen Müller, Torsten Hoefler

arXiv preprint, 2025 · [\[paper\]](#) · [\[code\]](#)

Higher-Order Graph Databases

Maciej Besta, **Shriram Chandran**, Jakub Cudak, Patrick Iff, Marcin Copik, Robert Gerstenberger, Tomasz Szydlo, Jürgen Müller, Torsten Hoefler

arXiv preprint, 2025 · [\[paper\]](#) · [\[code\]](#)

Maximus: A Modular Accelerated Query Engine for Data Analytics on Heterogeneous Systems

Marko Kabic, **Shriram Chandran**, Gustavo Alonso

10.1145/3725324, Proc. ACM Manag. Data (SIGMOD), 2025 · [\[paper\]](#) · [\[code\]](#)

PATENTS

System and Method for Automatic Parallel Code Generation for Graph Algorithms for Multiple Target Architectures

Rupesh Nasre, Ashwina Kumar, Ebenezer Rajadurai, Nibedita Behera, Rajesh Pandian Muniasamy, Shan Mohammed, Sai Nitish, Jash Khatri, Naveen S, **Shriram Chandran**, M. Venkatakrishna, Rushabh Lalwani

Indian Patent 432922, 2022 (Granted)

TECHNICAL EXPERTISE

Programming: C/C++, Python, CUDA, HIP/ROCm, Scala, Haskell, OCaml

Accelerator/ISA: PTX, x86-64, ARM64, RISC-V

Parallel/Distributed: MPI, OpenMP

Frameworks / Tools: Git, Linux, PyTorch

TEACHING AND MENTORING

Student Mentoring

ETH Zürich

- **Dominic Rinderer** *BSc Thesis, Spring 2026*
Research on automated CUTLASS kernel selection.
- **Nicholas Sommer** *MSc Thesis, Spring 2026*
Research on automatic generation of cuTile kernels.
- **Dalia Shaaban** *MSc Thesis, Autumn 2026*
Research on sparse workspaces for GPU sparse computations.
- **Dillon Elste** *MSc Thesis, Autumn 2026*
Research on automatic Tensor Core transformations.

Teaching Assistant

ETH Zürich

since Feb. 2026
Zürich, Switzerland

- 252-0029-00L Parallele Programmierung (Spring Semester 2026)
- 252-0836-00L Informatik II (Autumn Semester 2026)